REAL-TIME INTELLIGENT ROAD SAFETY MONITORING SYSTEM WITH VIOLATION DETECTION USING VIDEO SEGMENTATION

Project ID: RP 25-26J-470

Project Proposal Report

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BSc (Hons) in Information Technology Specializing in Software Engineering

Department of Computer Science and Software Engineering

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Road safety and public security remain pressing global challenges, with millions of lives affected annually by road accidents, violence, and weapon-related crimes. Despite advances in surveillance technology and Artificial Intelligence, current systems often depend on manual monitoring and reporting, resulting in late or insufficient responses. This research presents an intelligent system, specifically designed for road environments that leverages computer vision, deep learning, and mobile technologies to address these gaps by providing real-time, automated detection and intervention mechanisms. Primary module processes surveillance videos through sophisticated deep learning models, analyze video streams in real-time to detect accidents. And it simultaneously operates to detect violent activities, objects through hybrid deep learning approaches. All the incidents will be classified along with contextual factors that may contribute to escalating tensions. Subsequently, the system retrieves and processes data from all detection modules to create comprehensive situational awareness. It analyzes processed incident data to automatically determine which authorities should be notified based on the details, which includes comprehensive documentations featuring exact incident details. Upon detecting unsafe driving behaviors, shifts to a proactive accident prevention approach, i.e., a mobile app, through continuous monitoring of individual driving sessions. All these four interconnected layers – (1) accident detection layer, (2) hybrid violence detection layer, (3) central coordination hub, (4) driving assistant aims to address road safety by developing an integrated system that combines state-of-the-art detection algorithms with practical alert generation and response coordination capabilities, contributing to the advancement of intelligent public safety infrastructure. This research is expected to demonstrate that efficient detection and analytics can significantly improve real-time response to road-related incidents and enhanced public trust in intelligent transportation systems.

Keywords – Deep Learning, accident detection, YOLOv8, response times, violence detection, CNN

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INTRODUCTION

Road safety is a critical interdisciplinary field that addresses one of the most significant public challenges facing modern society. Each year, more than one million people die in road accidents, while another huge number suffer non-fatal injuries. This makes road safety one of the leading causes of death and disability across the globe. The problem is further exacerbated with the gap between incident occurrence and timely intervention. A significant portion of these emergencies go undetected or receive delayed responses specially like during off-peak hours. Similarly, road violence incidents - including assault, robbery, carjacking, and targeted attacks, often unfold rapidly with no immediate intervention mechanism in place. During these critical moments when help is most urgently needed, victims and bystanders find themselves isolated, vulnerable, and without access to immediate assistance or protection. So, such incidents present unique challenges which should escalate rapidly to ensure public safety.

Although subsequent decades have witnessed the integration of increasingly sophisticated methodologies, including epidemiological approaches, behavioral psychology insights, engineering solutions, responses to road incidents still heavily depends on manual monitoring of CCTV footage and public intervention. Therefore, proposed solution addresses these fundamental deficiencies by integrating leading-edge computer vision and machine learning techniques to transform manual monitoring of CCTV into an intelligent real time detection system. The primary component processes surveillance videos to detect risky driving behaviors extracting critical details such as vehicle identification, location, and time of occurrence, while it simultaneously detects violent activities, objects and suspicious behaviors on roadways. Upon detecting violations, the system collects all data to create comprehensive situational awareness. The system analyzes data to ensure that incidents reach the most appropriate response agencies without delay. Additionally, system uses a proactive approach to accident prevention through continuous monitoring of individual driving sessions i.e., a mobile application. By personalizing, the system creates individualized driver profiles that adapt to specific behavioral patterns and provide targeted interventions. The entire system is built on a scalable architecture capable of doing processing video feeds to generation of comprehensive response actions.

One essential component of this research is the “Hybrid Violence and Weapon Detection for Road Safety”, mainly capable of detecting violent objects, aggressive actions, and suspicious behaviors in real time through advanced artificial intelligence methodologies. Aspect of this

component is to address significant delays in threat identification as possible. The module includes a hybrid deep learning approach combining YOLO-based object detection and CNN–LSTM action recognition enabling both spatial and temporal feature extractions. Following, the system assigns confidence scores to detected events within the models itself during the detection process, which is later fused using logical and environmental context-aware rules that considers factors such as location type, time of day that consider situational context to enhance detection accuracy and produce comprehensive threat assessments. Eliminating false alerts while ensuring that genuine threats receive appropriate priority is accomplished through this fusion process. The resulting alerts are structured according to threat classification protocols, that categorize incidents based on severity level, and threat type, with other important details attached.

Through the integration of advanced deep learning techniques, intelligent data fusion, and automated response coordination, the system provides a practical, effective solution for enhancing road safety and reducing the impact of violent incidents on public roadways. The system's ability to operate continuously, process multiple threat types, and coordinate appropriate responses makes it an essential to protect road users seeking public safety.

Background And Literature Survey

Overview of the domain

The main domain of this specific research falls under an intersection of artificial intelligence (AI), and intelligent systems, with a strong focus on enhancing public safety and crime prevention. The proliferation of surveillance cameras across urban areas, highways, and public spaces has facilitated a significant transition from manual video monitoring to automated, intelligent video analytics systems, creating unprecedented opportunities for real-time threat detection and response. This research specifically focuses on three critical subdomains: violence detection on roadways, action recognition in traffic environments, and comprehensive public safety monitoring systems. Technological approaches within this domain predominantly utilize machine learning methodologies, Support Vector Machines (SVM), and various deep learning techniques for spatial and temporal analytics. Therefore, “Violence Detector” employs deep learning as its core technological framework to identify and classify violent actions.

Overview of the background

Violence incidents including street altercations, armed robberies, stabbing attacks, and firearm-related violence, occur with alarming unpredictability across public spaces such as roadways and transportation hubs. The spontaneous nature of these incidents, creates significant challenges for traditional security and emergency response systems, which still rely heavily on manual detection and reporting mechanisms. Particularly within the Sri Lankan context, normal people attempt independent intervention to such incidents due to the absence of timely official responses, often places additional individuals at risk and may inadvertently escalate dangerous situations.

Furthermore, a substantial proportion of violent incidents either remain undetected by surveillance teams or are reported with significant temporal delays, resulting in preventable injuries and fatalities. To systematically address these, the proposed component employs a fast object detection model i.e., YOLOv8, and a hybrid deep learning model for action recognition, i.e., CNN + LSTM model, which undergo integration through logical + context hybrid fusion method, that minimizes false positive results.

1. Object Detection
2. Violent Action detection
3. Confidence scores
4. Logic + Context fusing
5. Alert structuring

Contribution to the existing studies

Violent object and action detection has been a key focus in computer vision and intelligent surveillance. Early approaches relied on classical methods such as SIFT, SURF, and Histogram of Oriented Gradients (HOG), often combined with machine learning classifiers like Support Vector Machines (SVM), Logistic Regression, Random Forest, or Linear Discriminant Analysis (LDA) [1][2][3]. These techniques successfully detected violent actions in surveillance videos or extracted local features, but they were limited in scalability, speed, and their ability to classify complex scenarios beyond binary violent/non-violent labeling [4].

The emergence of deep learning has significantly advanced the field. Recurrent models such as LSTM have been applied for real-time violence detection by analyzing sequential video frames, but their temporal-only focus limits contextual understanding [5]. Convolutional

models like ResNet detect spatial features efficiently and are lightweight, though they may underperform in handling false positives/negatives [6]. Combining ResNet50 with ConvLSTM addresses both spatial and temporal aspects, yet scenario classification remains limited. Alternative bio-inspired approaches such as Spiking Neural Networks efficiently encode temporal features while remaining memory-effective, though they still lack detailed scenario classification [7].

Fusion strategies have been explored to combine object and action detection outputs, typically using classifiers like SVM. While datasets such as COCO, TAO, MoLa InCar, and XD-Violence support training, prior research often overlooks context-based fusion for enhanced scenario understanding [8][9].Some studies have used smaller datasets [13]. CNN + LSTM models achieve high accuracy in detecting violent actions [10], and some campus-focused systems even incorporate facial recognition [11], yet none fully integrate real-time object detection with action recognition and context-aware classification.

This research proposes a hybrid solution combining YOLOv8 for violent object detection and CNN + LSTM for violent action recognition. The system assigns confidence scores to each detection and applies a logic- and context-based fusion method to generate detailed, scenario-classified alerts. By combining spatial, temporal, and contextual analysis, this component aims to advance violent incident detection, providing faster, more accurate, and context-aware responses than existing approaches.

Research Gap

Table 1: Research Gap

Aspect	**Existing Systems (Research)**	**Proposed Application Component**
Classical Object Detection	Use of HOG + SVM, HOF, and classic ML models like LDA, Random Forest for violence/action detection [1][2]. SIFT/SURF used for key point detection in image manipulation [3].	Move from classical handcrafted features to deep learning-based models – YOLOv8/YOLOv11

Action Recognition	Human Action Recognition (HAR) to detect violence by classifying activities to binary [4]. Only one model - LSTM (for temporal features) [5], ResNet (for spatial features) [6], and ConvLSTM [7].	CNN+LSTM hybrid model- captures spatial and temporal features for richer behavior detection.
Fusion Strategy	Lightweight models like ResNet used for edge devices but lacking accuracy and false positive handling [6]. Spiking Neural Networks used for low-resource devices but limited in scenario classification [7].	Logic + context fusion - combines objects + actions + scene cues, which gives detailed classification
Output	Binary results (violent / non-violent) only [7]	Scenario-based, more detailed results
Dataset	Reliant on single datasets and datasets are specified like uses COCO + TAO (objects), MoLa InCar (actions) [8]	Dataset related to both object and actions are flexible, adaptable to varied security footage.
Real-Time Use	All the research studies lack alert generation part wrapped with better classifications	Detailed alerts preserved.

Prior research in this domain has primarily concentrated on developing isolated systems tailored for a single task, either object detection or action recognition focusing mainly on enhancing accuracy through model training alone. This approach, while effective within a narrow scope, often results in solutions that lack the context needed for real-world deployment. Such older systems can be prone to false alarms and may not provide the necessary understanding for critical decision-making. This research addresses these fundamental limitations by proposing and evaluating an innovative hybrid architecture that combines YOLOv8 and CNN-LSTM models through an intelligent logic-based fusion mechanism. Unlike previous approaches that rely solely on individual model performance, this approach integrates outputs from both detection models for accuracy. Consequently, the proposed solution will generate high-quality, reliable, and flexible alerts.

Research Problem

Urban road networks and public transportation corridors are experiencing unprecedented security challenges due to escalating violent incidents including armed assaults, vehicular attacks, road rage confrontations, gang-related violence, and organized criminal activities nowadays. Current surveillance infrastructure presents multiple critical limitations that compromise public safety and emergency response effectiveness.

Existing automated systems for violence detection using technologies like HOG, SVM, and even systems using machine learning approaches often focus on either object identification or action recognition in controlled conditions, but rarely address the human in need situations and process immediate action. Human operators demonstrate low accuracy rates in real-time threat identification, with response delays averaging 3-7 minutes for incident recognition and alert generation.

Previous violence detection systems utilizing traditional computer vision approaches (HOG, SVM, optical flow) and early-generation machine learning models exhibit significant deficiencies. Furthermore, these systems lack integration with rapid-response protocols, further compounded by the weak training of models trained on data. These studies lack multi-modal fusion architectures that effectively combine spatial object detection, temporal motion analysis, and contextual environmental data, resulting in poor performance in distinguishing between normal physical interactions and genuine violent threats.

Moreover, existing systems lack seamless integration with emergency services, law enforcement teams, and broader smart city infrastructure. Given the growing importance of public safety, there is a pressing need for a reliable, real-time system that can accurately detect violent incidents in road, minimize false alarms and to facilitate immediate emergency responses. Proposed system aims to address all these challenges by developing a hybrid detection model to detect violent object and actions, which eventually combines results, and seamless emergency service integration with classifying alerts reliably.

The system employs a hybrid detection approach, where YOLO-based object detection and CNN–LSTM models capture both spatial and temporal features, creating more accurate and reliable incident recognition. Detected events are then given confidence values, analyzed not only individually but also combined through confidence-based fusion strategies. These are

further processed with logical rules and enriched with contextual metadate such as time and location type. This multi-level fusion process produces structured and classified alerts that reflect the nature and severity of incidents, ready to send to central coordination hub for further processes.

OBJECTIVES

Main Objective

To develop and implement an intelligent real-time violence detection and emergency classification system that is capable of accurately detecting violent incidents in urban road networks. Unlike other systems which focus directly in detecting, this proposed solution aims to classify violent incidents again using fusion strategies to minimize false positives. Study aims to accurately identify objects using YOLOv8 and actions using CNN + LSTM hybrid model architecture, thereby enhancing public safety through timely and effective incident response.

Specific Objectives

- **Video Frame Extraction & Preprocessing:** Implement an efficient pipeline to extract and preprocess frames from live camera feeds or stored surveillance videos, ensuring optimal and correct input – per one frame for YOLOv8 and sequence of frames for hybrid model
- **Implement and fine-tune YOLOv8 model:** Train YOLOv8 model for accurate real-time detection of objects relevant to violent incidents such as knives, guns, blunt instruments, within urban traffic surveillance footage.
- **Design, Implement and train hybrid CNN+LSTM model:** Train and fine-tune a CNN + LSTM hybrid model to extract important spatial features from CNN, and understand temporal relationships of these spatial features through LSTM model. To achieve maximum accuracy in distinguishing between violent and non-violent activities in real-world traffic scenarios as possible.
- **Multi-Modal Fusion Integration:** Create a comprehensive fusion framework that combines confidence scores from video surveillance, with logical rules and contextual environmental data (time of the day, location type detected)
- **Alert Classification mechanism:** Upon detecting accurately, and successfully combining with contextual metadata and logic, system should classify alerts according to different severity levels and prioritize notifications – generating alert structures, for seamless integration with the central coordination hub.

METHODOLOGY

Understanding the Application of key pillars

The proposed *relies on principles of* computer vision, machine learning, and decision making. Therefore, proposed component draws upon principles of **pattern recognition, fusing accuracy,** and **decision making**

- **Pattern Recognition** – The component relies heavily on the detection results made by both object detection and action recognition models, As this is the first stage in the pipeline, high detection precision using patterns is essential to ensure the reliability of all subsequent processes. This includes correctly identifying small, fast-moving objects (e.g., hiding knives, quickly appearing guns) if possible and detecting violent behaviors with minimal error.
- **Fusing accuracy** – Even if the detection stage gives results well, the accuracy of the fusion plays determining the final output. Fusing basically involves combining the confidence levels form both models and applying additional logic-based rules to reduce false positives and false negatives, this stage ensures that the final results are reliable, and validated.
- **Decision Making** – Once the fused detection results are done, the system must make decisions to effectively classify and prioritize alerts for action. This section should ensure that alerts are structured in a clear, standardized, and actionable format. Proper decisions leas proper alert prioritization ensure understating high-risk are send immediately, while lower-priority alerts are handled accordingly.

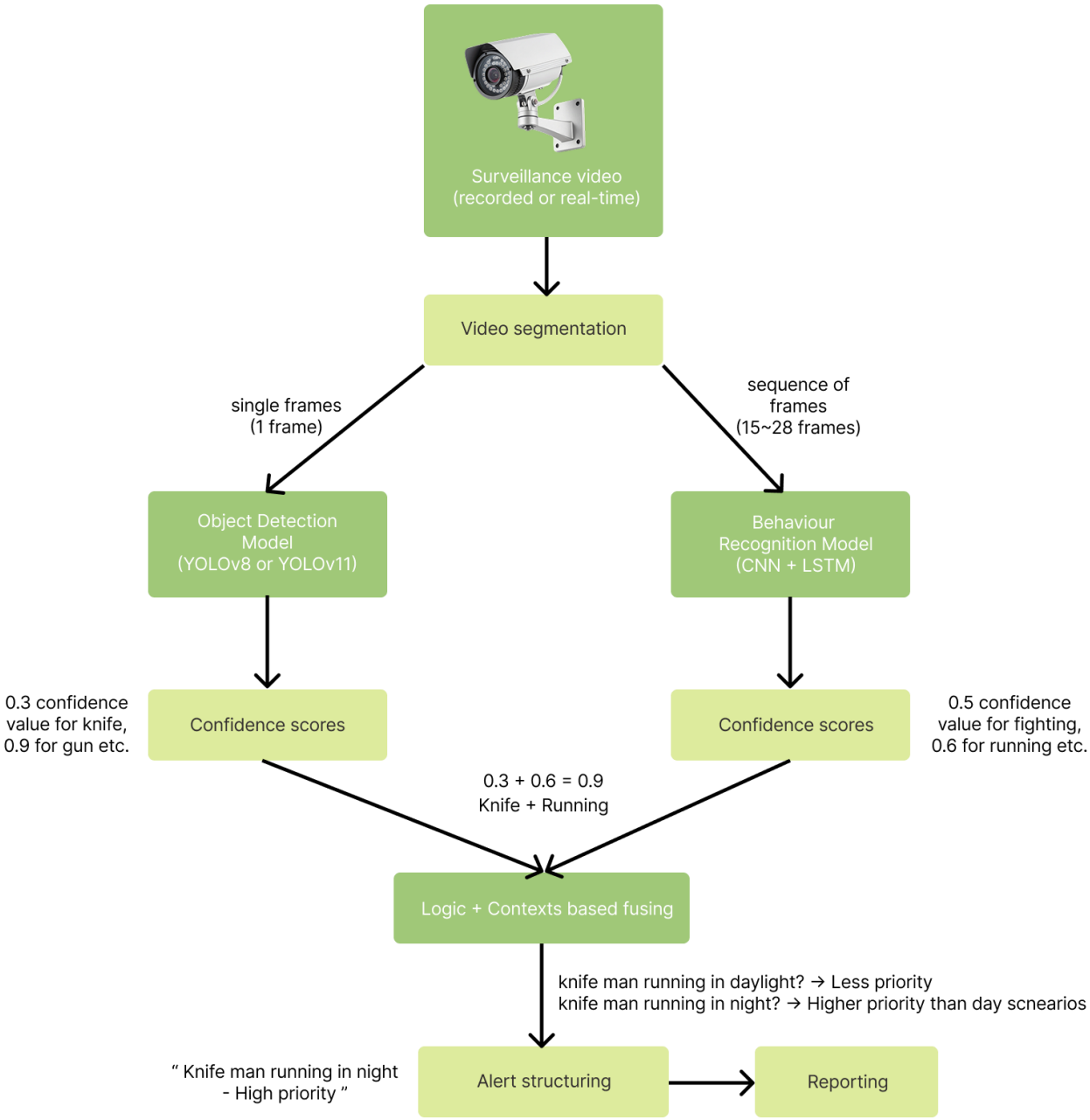

Figure 1: Violence Detector Component flow

System Architecture Diagram

The system diagram in this section illustrates how the four core components integrate to achieve the main objective of the research - AI-driven intelligent road safety monitoring system aiming enhancing road safety, reducing accident response times and promoting timely alerts. These components work collaboratively in a pipeline aligning with the research's goal

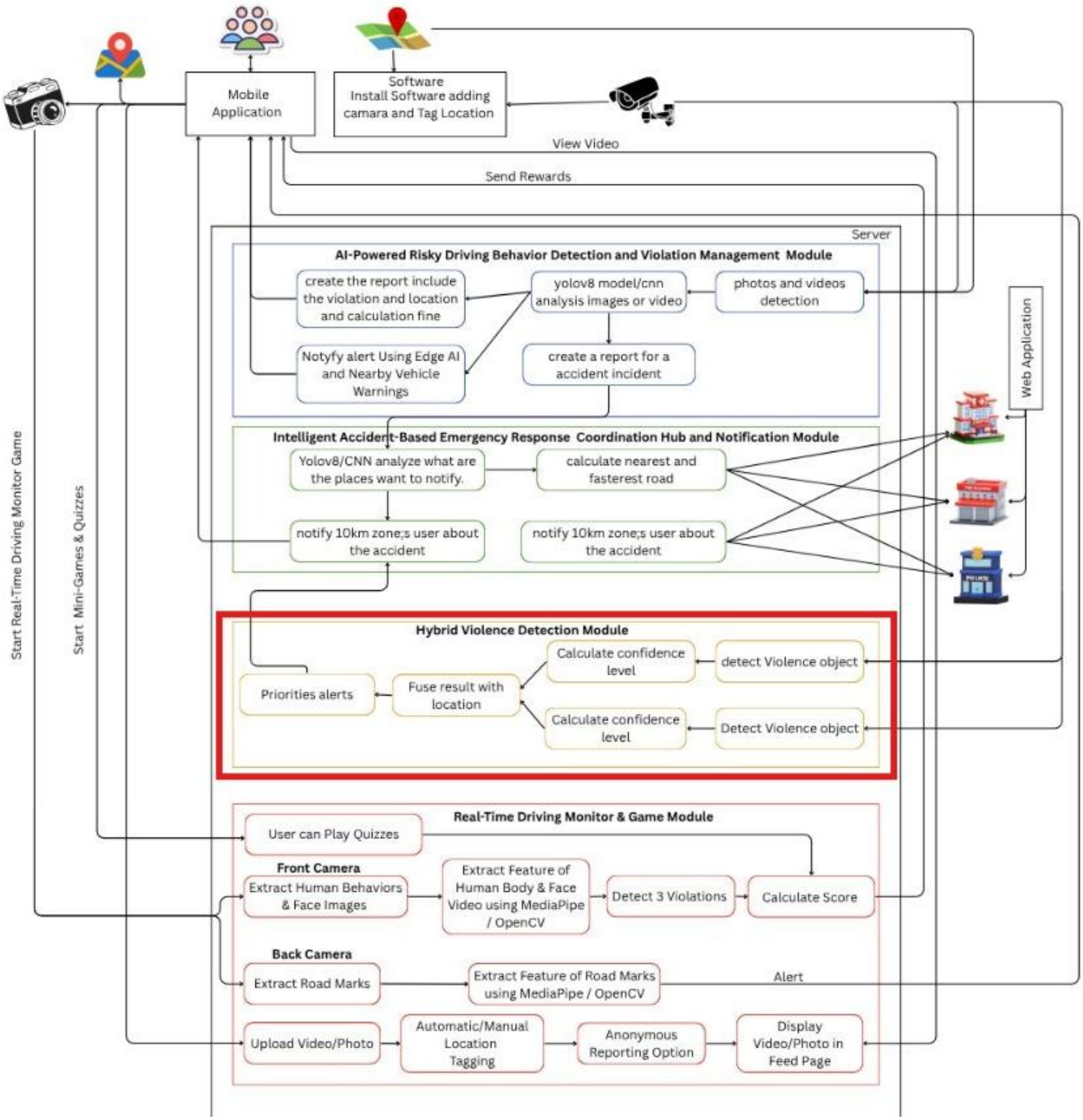

Figure 2: System Overview

Application of Technologies in the relevant area

For model development, this implementation uses manually provided or pre-recorded surveillance videos, so no hardware cameras are required, focusing solely on software-based processing.

- **Violent spatial and temporal feature detection:** Frameworks such as TensorFlow or PyTorch will be used to train and deploy YOLOv8 and CNN+ LSTM hybrid model. Custom dataset should be used to train both models.
- **Interface and Backend Server**: A lightweight and flexible frontend (React), and a strong heavy backend (Python Django/FastAPI for integration with models, Java Spring Boot/Node.js for handling heavy processing, feature development, and system logic). This architecture ensures efficient model inference, feature development while maintaining a clean separation between frontend and backend

Data Requirements and Collection Method

The project will require two main types of data

1. **Object detection datasets** - for identifying potential weapons or hazardous objects
2. **Action recognition datasets -** for detecting violent activities or suspicious behaviors

For object detection it needs datasets containing labeled images and videos of knives, guns, and other dangerous items will be sourced from public datasets – Roboflow. For action recognition, video datasets showing various violent and non-violent scenarios will be used. No direct survey or interview will be required since the project is based on video data collection and annotation. However, if integration with emergency services will need external support and data, interviews with such personnel is conducted frequently to gather operational requirements.

Project Execution

Project execution will carry on in several iterative development processes.

Phase 1 - Training Data Preparation for Models

- Identify datasets for training (violent object datasets, violence action datasets)
- Prepare object detection datasets for YOLO training.
- Prepare action recognition datasets for CNN+LSTM training.

Phase 2 - Model Development

- Train a YOLO model for violent object detection.
- Train a CNN+LSTM model for detecting violent actions.

Phase 3.1 - Develop Video Input section and Model Integration

- Implement method to insert real-time video or recorded videos to the module
- Implement frame extractions (single frames and series of frames)
- Implement how 2 models combine with real time videos to detect

Phase 3.2 – Implementing Confidence scores and Ready to Fuse

- After results should be given confidence scores

Phase 4 – Fusion Methodology and Alert structuring

- Implement logic-based fusion to give logic based responses
- Implement context based fusion to give environmental based guessing based responses
- Combine all to generate alert structures
- Ready to integrate and send to central hub

Phase 5 - Testing and Validation

- Perform system testing for surveillance feeds
- Evaluate accuracy of the models, and precision of the results
- Evaluate speed of generating final responses
- Ensure successful integration into full Intelligent Road Safety Monitoring System

Project requirements

Functional requirements

1. **Detect and classify violent objects (YOLO)**
   - The module receives video input from the designated source and processes it by breaking the footage into individual frames. Each frame is analyzed to identify objects associated with violent activities (e.g., knives, guns).
   - Confidence scores should be assigned based on the identified objects (e.g: 0.2 for a knife, 0.5 for a gun)
   - All evidence videos and related details should be stored securely for analysis and reporting.
2. **Recognize suspicious or violent human actions (CNN+LSTM)**
   - The module processes the received video input from the designated source and should implement it to process sequences of video frames to detect abnormal or violent actions (e.g: fighting, hitting, kicking)

- Should determine if accurate results are generated with CNN extracts spatial features (from each frame), and LSTM capture the temporal motion patterns across previous frames.
- Confidence scores should be assigned based on the identified objects (e.g: 0.9 for fighting for a knife, 0.3 for running behavior)
- Response details should be stored securely for analysis and reporting

3. **Fusing results**
   - The confidence scores generated from the above models will be combined using a logic fusion method, which compares the outputs from both models to produce a final result.
   - This process also incorporates fusing the logic with contextual metadata(e.g: time, location), enabling logic-based comparisons over these – e.g: daylight loitering is considered low impact, whereas nighttime loitering is considered high impact.
4. **Generate threat alert structures**
   - The final results are processed to generate user-friendly alert structures, designed to be easily understood
   - These alert structures, along with the detailed data stored from the models, should be prepared for transmission to the next component - the central hub.

User Requirements

- **Alert details generated**

  The system should automatically generate and forward alerts to the central hub as soon as a violent incident is detected, ensuring minimal delay. Should provide complete details allows authorities to quickly understand the context of the event and determine the appropriate course of action.
- **Detail analytics**

  It can provide additional features such as incident reporting, displaying violent incident analytics and presenting video evidence for review, for the user interface, easier access to actionable insights for users.

System Requirements

- **High computational performance**

  The system should be capable of handling complex computations with minimal latency, ensuring that incident detection are processed fast. This requires powerful processors, sufficient memory, and GPU acceleration for efficient process.
- **Compatibility with video input formats and quality in detecting**

The system must accept all video formats and videos from different input methods in different qualities. Should ensure seamless detection quality detection with different camera angles with different video qualities.

- **System architecture for large deployments including models**

  The architecture should be designed to support development using standard programming languages such as Java, Python, or JavaScript, which python is for model development and deployments. It must be scalable to handle enterprise-level deployments, allowing distributed processing and integration with other components

Non-Functional Requirements

- **Accuracy** -The violence detection module must achieve at least 85% accuracy in identifying violent objects and actions to minimize false alarms.
- **Performance** - The two models should perform well in order to process video frames accurately.
- **Reliability** -Reliability of the results produced from fusing.
- **Usability** -Alerts should be well organized and classified
- **Security** -Sensitive data, including video streams and alert information, must be handled securely

Commercialization aspects of the project

Target Audience for the component:

- Law Enforcement Authorities - Police Departments
- Government Agencies – Hospitals, Fire and Rescue Services
- Smart City Administrators
- Transportation Authorities
- Emergency Response Coordinators

Market Space

1. Law Enforcement Authorities - Police departments needing automated surveillance tools to detect violent activities, improve response times
2. Government Agencies **-** Including hospitals, fire units, and other emergency services that benefit from timely alerts and timely incident management.
3. City Infrastructure **-** Municipalities using deep Learning AI driven surveillance to monitor roads which capture public areas
4. Private Security firms - Companies managing security for universities, schools, businesses for their use

Anticipated Conclusion

The Violence Detector component is expected to significantly enhance real-time road safety monitoring by first, accurately detecting and classifying violent incidents using advanced deep learning models, second combining results with context + logic aware fusion to logic based decisions about incidents, finally generate alert structures classified with priority aiming faster and more informed responses and records.

DESCRIPTION OF PERSONAL & FACILITIES

The contributions of the whole project include a collaborative approach involving three sequential components and one public-use module. De Silva K.W.S.C (IT22060358) implements the first component, which captures surveillance videos, and detect and extract critical details such as illegal road behaviors to send to central hub component. D.P.S Ranasinghe (IT22105134) focuses on detecting violent behaviors and violent objects on the road, preparing this data to be sent to the central hub as well. Senavirathna K.T.M.K (IT22059918) is responsible in building the Emergency Response Coordination Hub (central hub which manages, analyzes, and decides which authority should be notified based on the incident type. Finally, Welandagoda D.R.L (IT22081902 focuses on the interactive module for public use, the mobile application that monitors driving sessions in real-time and provides live feedback to drivers

Facilities

The project leverages the university's computing infrastructure, including high-performance workstations with GPUs to support intensive model training, testing, and inference. Video datasets are processed using simulated video streams to replicate real-world environments under varying conditions and qualities. Core software tools include Python-based machine learning frameworks such as TensorFlow and PyTorch, and containerization platforms (Docker) for deployment. Cloud services are utilized for scalable model hosting, evaluation. This integrated setup ensures reliable performance, scalability, and robustness, supporting the development of the proposed system.

Expected Test Cases

Table 2: Expected Test Cases

Test Area	Test Case ID	Description	Input	Expected output
1. Input video streams	TC1-01	Testing viode input format compatibility	Input videos in MP4, and MKV – different formats	All formats accepted without errors, frames extracted correctly

	TC1-02	Different Qualities	Input videos with different resolutions	System adapts to different resolutions
	TC1-03	When testing with real camera feed	Live video stream	System processes frames with no visible lags
2. Object Detection Model Training	TC2-01	Positive Identification- correctly identifies relevant objects	single image clearly containing a handgun	class: 'gun' and confidence > 0.7.
	TC2-02	Negative Identification – doesn't identify irrelevant objects as weapons	A single static image of a peaceful scene	pre-generated short clip (16 frames) of
Action detection model training	TC3-01	Positive Identification - correctly classifies a violent action sequence	pre-generated short clip of a fighting action	"violent" class, confidence > 0.7
	TC3-02	Negative identification – identifies non-violent action successfully	pre-generated short clip of	List is empty or confidence scores < 0.3
Fusion Logic	TC3-01	Consistency Check	YOLO detects object + CNN classifies incident type	Final decision matches details
	TC3-02	Conflict Resolution	YOLO detects- "people fighting," CNN - "crowd gathering.	"fight" classification
Alert Generation	TC4-01	Reliable alert generation	Outputs form the fusing mechanism	A structured alert format suitable to send to central hub

BUDGET AND BUDGET JUSTIFICATION

The estimated budget for this component primarily covers resources and costs related to software tools. However, if model training is conducted on free notebooks such as Google Colab and Jupiter Notebook, instead of premium versions, the training cost will be zero. Other expenses include minor costs such as internet connectivity and software licenses. Therefore, the overall main budget for the component remains minimal and focused mainly on supporting essential software access and connectivity required for the research. Considering the whole study, other components cover resources and costs related to hardware and software platforms. Below is a clear summary and a compact budget table considering whole research and research period.

Table 3: Expected Budget

Requirement	Planned Budget / Cost
AWS EC2 GPU Instance (g4dn.xlarge) – Cloud model hosting for YOLOv8 inference	$0.526/hour (on-demand, only during active sessions)
AWS S3 – Short-term event clip storage (≤30 days)	$0.023/GB (first 5 GB free per month)
AWS Route 53 – Domain & DNS management	$12/year
AWS Certificate Manager – SSL/TLS certificates	Free
Redis Cloud – Real-time session management	Free tier (30 MB storage, 30 connections)
MongoDB Atlas – Trip data storage	Free tier (500 MB storage)
Flutter Development Tools – Cross-platform app development	Free
Local GPU Workstation – Prototype testing & model training	Existing hardware (no cost)

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APPENDICES

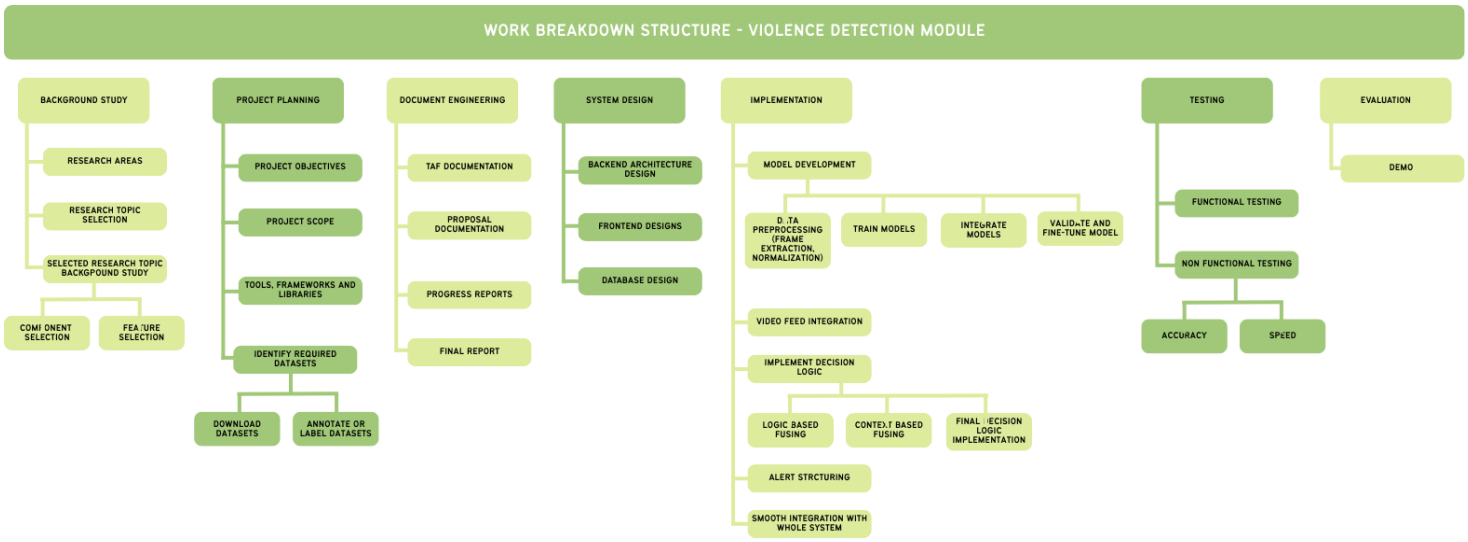

Figure 3: Work Breakdown Structure

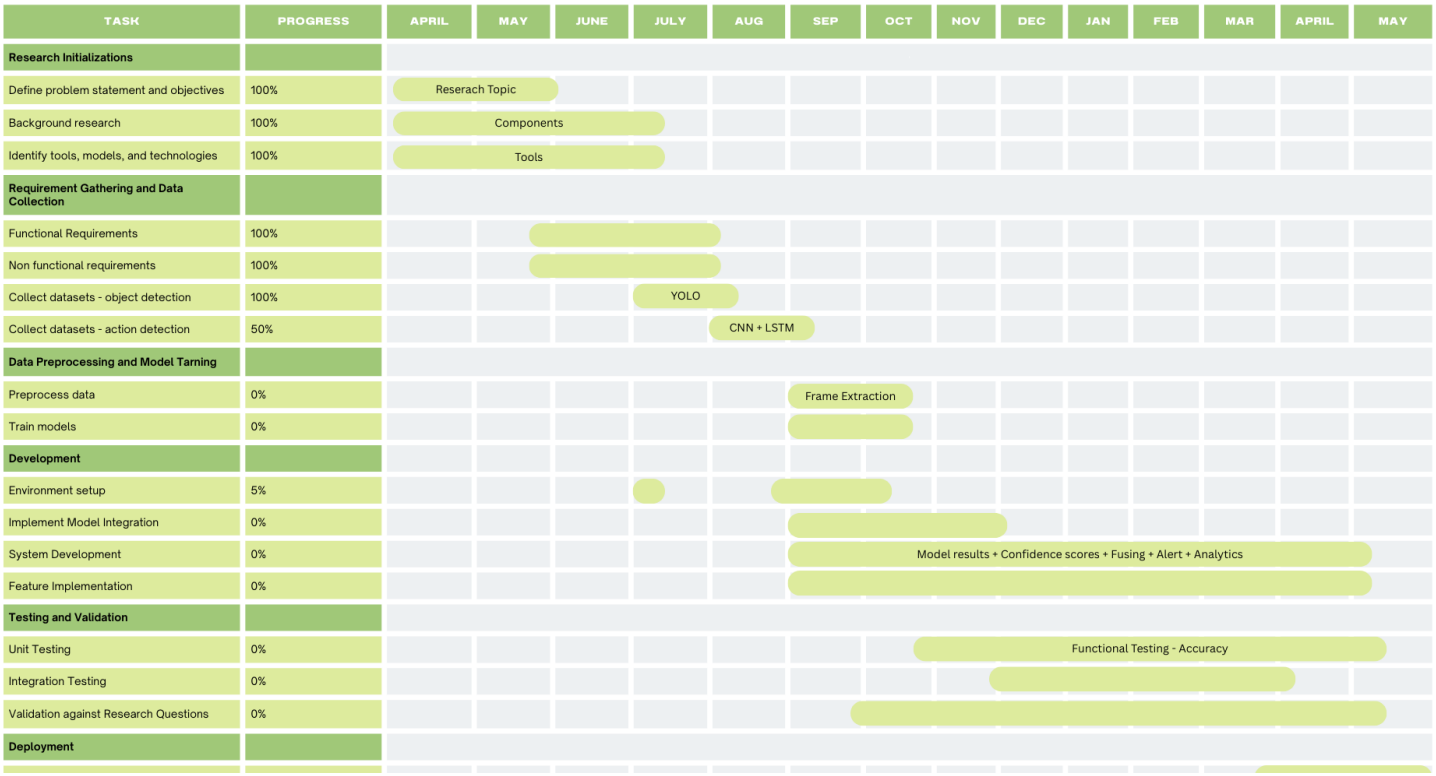

Figure 4:Gannt Chart